Tutorial- 7

1. Consider the classic Prisoner's Dilemma game with the following payoff matrix:

	Prisoner 2: Cooperate	**Prisoner 2: Defect**
Prisoner 1: Cooperate	(-1, -1)	(-10, 0)
Prisoner 1: Defect	(0, -10)	(-5, -5)

   Suppose this game is played for a finite number of rounds, say 3 times. Both players know that the game will end after the third round. What is the subgame perfect Nash equilibrium of this finitely repeated game? Explain your reasoning using backward induction.

2. In a finitely repeated game, a "grim trigger" strategy is one where a player cooperates as long as the other player cooperates, but if the other player defects even once, the first player will defect for the rest of the game. Consider the same Prisoner's Dilemma game from Question 1, repeated for 5 rounds. Is a strategy profile where both players adopt a grim trigger strategy a Nash equilibrium? Why or why not?

3. In repeated games, the discount factor, denoted by δ (where $0< \delta<1$), represents how much players value future payoffs relative to present payoffs. A higher δ means players are more patient. Consider a finitely repeated game with a known end, where the stage game has a unique Nash equilibrium that is not Pareto optimal (like the Prisoner's Dilemma). How does the value of the discount factor affect the possibility of cooperation in this game? Does